

BLA-FedMoE: Bayesian Layer-Wise Aggregation for Federated Learning with Mixture-of-Experts Large Language Models

Anonymous submission

Abstract

Federated learning (FL) has emerged as a promising privacy-preserving paradigm for training large language models (LLMs). To overcome the prohibitive parameter scale that hinders LLM deployment on resource-constrained edge devices, sparse mixture-of-experts (MoE) architectures are increasingly adopted. However, under highly non-IID data distributions, standard aggregation frameworks naively average expert parameters, which erodes their ability to capture local data characteristics and leads to severe performance degradation. To address this challenge, we propose a bayesian layer-wise aggregation algorithm for federated learning with MoE, named **BLA-FedMoE**. Specifically, the proposed method infers an approximate posterior distribution for each expert layer to quantify the parameter certainty regarding local heterogeneous data. Subsequently, we transform this certainty into a layer-level confidence matrix to guide adaptive parameter aggregation. Extensive experiments demonstrate that BLA-FedMoE consistently outperforms state-of-the-art baselines in both global test accuracy and convergence speed.

Introduction

Large language models (LLMs) have demonstrated remarkable generalization and reasoning capabilities across a broad spectrum of natural language processing tasks (Brown et al. 2020; Chen et al. 2024). However, their heavy reliance on massive public corpora poses significant deployment barriers in privacy-sensitive domains, such as healthcare and finance. Federated learning (FL) offers a promising decentralized paradigm to bridge this gap by enabling collaborative model training without sharing raw client data (Lin et al. 2026; Shi et al. 2024). Despite its privacy benefits, conventional FL requires clients to perform full-model local training and parameter updates. Given the massive parameter scale of modern LLMs (e.g., Llama-3 with 405 billion parameters and Mixtral 8×22B with 141 billion parameters), deploying these models on resource-constrained edge devices is practically infeasible.

The mixture-of-experts (MoE) architecture (Shazeer et al. 2017; Lepikhin et al. 2021; Xie et al. 2024) emerges as a compelling solution to this bottleneck. By employing a sparse

gating mechanism, MoE activates only a small subset of expert networks for each input token, effectively decoupling the computational overhead from the total model capacity. This sparse activation allows edge devices to participate in FL by updating only a fraction of the experts during training, thereby maintaining a large global model capacity while substantially reducing local memory and computational requirements. Consequently, integrating MoE with FL establishes a robust paradigm that concurrently preserves data privacy and overcomes the resource bottlenecks inherent in edge-based LLM training.

However, existing federated MoE frameworks still suffer from significant theoretical limitations and performance bottlenecks during the expert aggregation phase. Most existing methods directly adopt the classical FedAvg algorithm (McMahan et al. 2017; Tang, Wang, and Chang 2024) to average local expert weights, e.g., FedJETs (Dun et al. 2023) averages the updated selected experts and gating parameters, while pFedMoE (Yi et al. 2026) aggregates the uploaded homogeneous feature extractors through sample-size-weighted averaging. In highly non-IID settings, local data skew drives the experts trained on each client to capture features specific to its own data distribution. Such blind global averaging can impair the ability of expert networks to discriminate local features, thereby leading to performance degradation. Moreover, existing aggregation schemes are usually performed at the model level, neglecting the varying sensitivity of different neural network layers to data heterogeneity. Specifically, shallow layers tend to extract general low-level features, whereas deeper layers are more strongly coupled with local data characteristics. Therefore, due to the lack of mechanisms for modeling this inter-layer sensitivity, current methods fail to assign adaptive confidence weights to expert parameters across varying depths, resulting in performance degradation of the global model under non-IID conditions.

To address the above challenges, we propose a bayesian layer-wise aggregation algorithm for federated learning with MoE, named **BLA-FedMoE**. Unlike conventional parameter averaging methods, BLA-FedMoE adopts a Bayesian posterior inference perspective to model inter-client data heterogeneity and intra-expert layer-wise sensitivity. Specifically, the method first infers the posterior distribution for each layer of the expert networks during local training, thereby

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characterizing the local uncertainty along different parameter directions. Then, the server constructs a quadratic optimization objective based on the layer-wise posterior information uploaded by clients and iteratively solves for the global parameters of the corresponding expert layers. This mechanism allows expert layers that contribute more significantly to local feature capture to play a more dominant role in global aggregation, thereby effectively mitigating the pitfalls of layer-agnostic averaging in federated MoE models. The main contributions of this paper are summarized as follows:

- We propose BLA-FedMoE, the first layer-wise posterior inference-based aggregation framework for federated MoE. It provides a rigorous mathematical foundation to resolve parameter conflicts induced by non-IID data, preventing expert capacity erosion caused by naive averaging.
- By employing block-diagonal empirical Fisher approximations, we reformulate the high-dimensional posterior inference into a convex quadratic problem, which can be efficiently optimized with guaranteed linear convergence.
- Extensive experiments demonstrate that BLA-FedMoE significantly outperforms existing state-of-the-art model aggregation baselines under highly skewed data distributions.

The remainder of this paper is organized as follows. The related work section reviews existing studies on FL and MoE models. The methodology section presents the detailed algorithmic design of our method. The experiments section reports the experimental settings and performance evaluation. Finally, the conclusion section summarizes the paper.

Related Work

Federated Learning under Non-IID Data

FedAvg (McMahan et al. 2017; Tang, Wang, and Chang 2024) establishes the basic parameter aggregation paradigm in FL. However, under statistically heterogeneous, or non-IID, data distributions, divergent local objectives may cause client drift and degrade aggregation performance. Existing methods alleviate this issue through different mechanisms: FedProx (Li et al. 2020a) constrains local updates with a proximal term, SCAFFOLD (Karimireddy et al. 2020) corrects client drift using control variates, MOON (Li, He, and Song 2021) adopts model-level contrastive regularization, FedDF (Lin et al. 2020) performs server-side knowledge distillation, and FedPer (Arivazhagan et al. 2019) separates shared and personalized layers. Despite their effectiveness, these methods mostly rely on complete client models and coarse-grained model- or layer-level aggregation, limiting their ability to capture fine-grained client-specific patterns. This motivates us to explore FL with sparsely activated MoE architectures for more flexible expert specialization under statistical heterogeneity.

Federated Learning with Mixture-of-Experts

Several recent studies have explored FL with MoE architectures. FedMix (Reisser et al. 2021) maintains and aggregates

expert subsets across clients to enable multi-domain knowledge fusion. FedJETs (Dun et al. 2023) aims to reduce communication overhead by suppressing the upload of less informative experts. pFedMoE (Yi et al. 2026) separates global feature extractors from local personalized experts and fuses their representations for local adaptation. FedMoE-DA (Zhan et al. 2024) uses gating parameters as routing priors to match semantically corresponding experts on the server. Despite these advances, expert aggregation in FL with MoE remains insufficiently explored. Under highly non-IID data, uniform parameter averaging may blur expert-specific knowledge and weaken local feature specialization, motivating a more reliable expert aggregation framework.

Methodology

Federated Learning with MoE from a Posterior Perspective

FL is a collaborative distributed training paradigm in which multiple clients jointly optimize a global model without sharing raw data. Given K participating clients, the global objective is to minimize the following weighted empirical risk:

$$\min_{\theta} f(\theta) = \sum_{k=1}^K \frac{n_k}{n} f_k(\theta), \quad (1)$$

where $n = \sum_{k=1}^K n_k$ denotes the total data volume across all clients, n_k is the local data volume of client k , and $f_k(\theta)$ represents the empirical loss computed on local dataset $\mathcal{D}_k$.

While Eq. (1) describes the standard global optimization objective in FL, in our MoE framework, the local model is composed of multiple expert networks and a gating mechanism. For each input sample, the gating mechanism dynamically routes its feature representation to the designated expert networks. Therefore, the local loss of client k can be further decomposed into the expected sum of losses computed by all local experts on their corresponding data subsets:

$$f_k(\theta) = \sum_{e=1}^E f_k^{(e)}(\theta^{(e)}), \quad \mathcal{D}_k = \bigcup_{e=1}^E \mathcal{D}_k^{(e)}, \quad (2)$$

where E denotes the number of expert networks per client, and $\mathcal{D}_k^{(e)}$ is the data subset assigned to the e -th expert.

Minimizing the local expert loss $f_k^{(e)}(\theta^{(e)})$ yields a point estimate (Bishop and Nasrabadi 2006), which does not explicitly capture the local geometry of the loss landscape. Under non-IID data distributions, client experts may converge to regions with different curvatures, reflecting different levels of parameter confidence (Karimireddy et al. 2020; Liu et al. 2024). Naive parameter averaging ignores such curvature-induced confidence differences and may degrade the global expert. Therefore, we reinterpret the local expert objective from a probabilistic inference perspective, where $f_k^{(e)}(\theta^{(e)})$ can be formulated as the negative log-likelihood of the expert-routed local data subset $\mathcal{D}_k^{(e)}$ under parameters $\theta^{(e)}$:

$$f_k^{(e)}(\theta^{(e)}) = -\log p(\mathcal{D}_k^{(e)} | \theta^{(e)}). \quad (3)$$

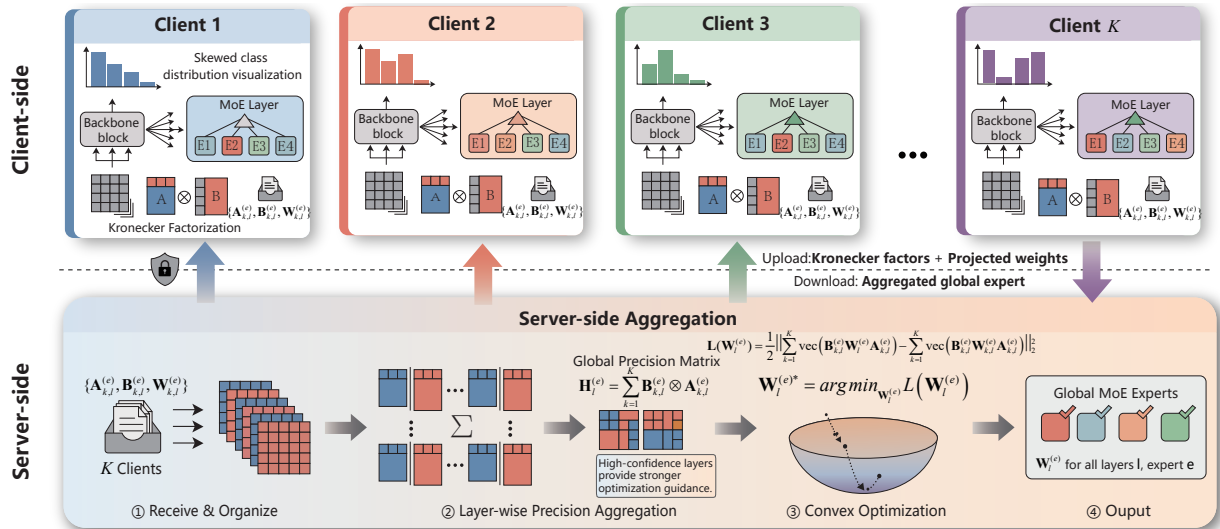


Figure 1: Overview of the proposed BLA-FedMoE framework.

This indicates that minimizing $f_k^{(e)}(\theta^{(e)})$ in federated MoE corresponds to local maximum likelihood estimation. However, such a likelihood-only formulation relies solely on the expert-routed local data and yields a point estimate without explicitly modeling parameter uncertainty. To address this limitation, we introduce a prior distribution $p(\theta^{(e)})$ over the parameter space. By Bayes' theorem, the local objective can be reformulated as posterior inference over $\theta^{(e)}$:

$$p(\theta^{(e)}|\mathcal{D}_k^{(e)}) = \frac{p(\mathcal{D}_k^{(e)}|\theta^{(e)}) \cdot p(\theta^{(e)})}{p(\mathcal{D}_k^{(e)})}. \quad (4)$$

Within the federated MoE framework, let $\mathcal{D}^{(e)} = \mathcal{D}_k^{(e)} k = 1^K$ denote the expert-routed data associated with expert e across clients. Following the standard conditional independence assumption in FL, these client-wise data subsets are independent given the expert parameters $\theta^{(e)}$. Therefore, up to a normalization constant, the global posterior over $\theta^{(e)}$ can be factorized into the prior and client-wise likelihoods:

$$p(\theta^{(e)}|\mathcal{D}^{(e)}) \propto p(\theta^{(e)}) \prod_{k=1}^K p(\mathcal{D}_k^{(e)}|\theta^{(e)}). \quad (5)$$

These derivations show that the aggregation problem for FL with MoE can be formulated as maximizing the global posterior in Eq. (5). However, for deep expert networks, the marginal likelihood $p(\mathcal{D}_k^{(e)})$ in Eq. (4) and the client-wise likelihood terms in Eq. (5) are difficult to compute exactly. We therefore introduce an empirical Fisher-based layer-wise approximation to obtain tractable precision estimates for expert aggregation.

Layer-Wise Posterior Approximation with Empirical Fisher

To avoid the prohibitive cost of full-posterior approximation while preserving the layer-wise sensitivity of expert

networks (Ritter, Botev, and Barber 2018), we approximate the posterior of same-index experts at the layer level across clients. Since the exact posterior is intractable, we adopt a Laplace-style approximation by treating the locally trained parameters as the local posterior mode and applying a second-order Taylor expansion around this mode:

$$\log p(\theta_{k,l}^{(e)}|\mathcal{D}_k^{(e)}) \approx \log p(\theta_{k,l}^{(e)*}|\mathcal{D}_k^{(e)}) - \frac{1}{2} \Delta \theta_{k,l}^{(e)\top} \mathbf{H}_{k,l}^{(e)} \Delta \theta_{k,l}^{(e)}. \quad (6)$$

where $\Delta \theta_{k,l}^{(e)} = \theta_{k,l}^{(e)} - \theta_{k,l}^{(e)*}$, $\theta_{k,l}^{(e)*}$ is the local optimum of layer l in expert e for client k , and $\mathbf{H}_{k,l}^{(e)}$ denotes the corresponding negative Hessian. Under the Laplace approximation, the local layer-wise posterior is approximated by a Gaussian distribution obtained from the second-order expansion around the local posterior mode. For the data subset $\mathcal{D}_k^{(e)}$ routed to the e -th expert, the post-training layer- l parameters $\theta_{k,l}^{(e)}$ are used as the mean $\mu_{k,l}^{(e)}$. The corresponding precision matrix, defined as the inverse covariance matrix, captures the local curvature of the negative log-posterior and thus reflects the confidence of the layer-wise expert parameters. $\mathbf{H}_{k,l}^{(e)}$:

$$p(\theta_{k,l}^{(e)}|\mathcal{D}_k^{(e)}) \approx \mathcal{N}(\theta_{k,l}^{(e)} | \mu_{k,l}^{(e)}, (\mathbf{H}_{k,l}^{(e)})^{-1}). \quad (7)$$

According to the conjugate property of Gaussian distributions, the global posterior of a specific expert layer remains Gaussian. Therefore, the global precision matrix and the global parameter expectation can be derived as:

$$\mathbf{H}_l^{(e)} = \sum_{k=1}^K \mathbf{H}_{k,l}^{(e)}, \quad (8)$$

$$\mu_l^{(e)} = (\mathbf{H}_l^{(e)})^{-1} \sum_{k=1}^K \mathbf{H}_{k,l}^{(e)} \mu_{k,l}^{(e)}. \quad (9)$$

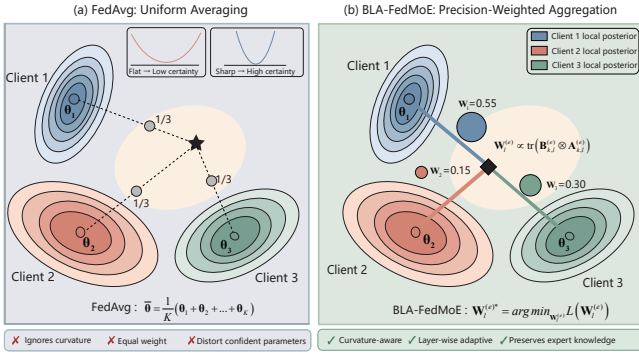


Figure 2: Comparison of FedAvg and BLA-FedMoE aggregation.

However, the precision matrix in Eq. (7) is high-dimensional, and the matrix inversion in Eq. (9) is computationally infeasible. Under the Laplace approximation, the second-order curvature of the parameter space can be approximated by the variance of first-order gradient fluctuations. Therefore, we approximate the precision matrix with the empirical Fisher information matrix, replacing second-order derivative computation with first-order gradient outer products to reduce computational and storage costs. Specifically, from Eq. (4) and Eq. (6), we have:

$$\log p(\theta_{k,l}^{(e)} | \mathcal{D}_k^{(e)}) = \log p(\mathcal{D}_k^{(e)} | \theta_{k,l}^{(e)}) + \log p(\theta_{k,l}^{(e)}) - \log p(\mathcal{D}_k^{(e)}). \quad (10)$$

Taking the second derivative of both sides with respect to $\theta_{k,l}^{(e)}$:

$$\mathbf{H}_{k,l}^{(e)} = -\nabla_{\theta}^2 \log p(\mathcal{D}_k^{(e)} | \theta_{k,l}^{(e)}) - \nabla_{\theta}^2 \log p(\theta_{k,l}^{(e)}). \quad (11)$$

By the Fisher information identity, the expected negative Hessian of the log-likelihood can be approximated by the expectation of first-order gradient outer products. Thus, we use the empirical Fisher matrix to capture local curvature without explicitly computing second-order derivatives. Since a Gaussian prior corresponds to standard weight decay regularization (Kirkpatrick et al. 2017), we assume an approximately Gaussian prior. Under this assumption:

$$\mathbf{H}_{k,l}^{(e)} \approx \mathbf{F}_{k,l}^{(e)} + \lambda \mathbf{I}, \quad (12)$$

where $\mathbf{F}_{k,l}^{(e)}$ denotes the empirical Fisher information matrix, and $\lambda \mathbf{I}$ is a constant diagonal matrix corresponding to the prior precision. To achieve computational and storage efficiency, we adopt a layer-wise Kronecker factorization for the empirical Fisher matrix. By invoking the statistical independence between forward activations and backward gradients, the Kronecker decomposition avoids explicit inversion of high-dimensional matrices and compresses the communication burden from a full matrix to two low-dimensional factors. We factorize the empirical Fisher as the Kronecker product of an activation covariance matrix $\mathbf{A}$ and a gradient covariance matrix $\mathbf{B}$:

$$\mathbf{F}_{k,l}^{(e)} \approx \mathbf{B}_{k,l}^{(e)} \otimes \mathbf{A}_{k,l}^{(e)}, \quad (13)$$

where $\otimes$ denotes the Kronecker product, and $\mathbf{A}_{k,l}^{(e)} = \mathbb{E}[\mathbf{a}_{k,l}^{(e)}(\mathbf{a}_{k,l}^{(e)})^\top] + \gamma_A \mathbf{I}$ and $\mathbf{B}_{k,l}^{(e)} = \mathbb{E}[\mathbf{g}_{k,l}^{(e)}(\mathbf{g}_{k,l}^{(e)})^\top] + \gamma_B \mathbf{I}$, $\mathbf{a}_{k,l}^{(e)}$ is the forward input activation received at layer l , $\mathbf{g}_{k,l}^{(e)}$ is the backpropagated gradient vector with respect to the output of layer l , and γ_A, γ_B are damping coefficients. Thus, the layer-wise local posterior of each expert is approximated as:

$$p(\theta_{k,l}^{(e)} | \mathcal{D}_k^{(e)}) \approx \mathcal{N}(\mu_{k,l}^{(e)}, (\mathbf{B}_{k,l}^{(e)} \otimes \mathbf{A}_{k,l}^{(e)} + \lambda \cdot \mathbf{I})^{-1}). \quad (14)$$

Complexity Analysis. If the empirical Fisher matrix is directly constructed, then for a fully connected layer with input dimension d_{in} and output dimension d_{out} , the corresponding empirical Fisher matrix has a size of $(d_{\text{in}} \cdot d_{\text{out}}) \times (d_{\text{in}} \cdot d_{\text{out}})$. Storing this matrix requires a space complexity of $\mathcal{O}(d_{\text{in}}^2 \cdot d_{\text{out}}^2)$, while the subsequent matrix inversion required for aggregation further incurs a computational complexity of $\mathcal{O}(d_{\text{in}}^3 \cdot d_{\text{out}}^3)$. After applying Kronecker factorization, only two compact matrices, $\mathbf{A}_{k,l}^{(e)}$ and $\mathbf{B}_{k,l}^{(e)}$, need to be stored, with sizes $d_{\text{in}} \times d_{\text{in}}$ and $d_{\text{out}} \times d_{\text{out}}$, respectively. As a result, the storage complexity is reduced to $\mathcal{O}(d_{\text{in}}^2 + d_{\text{out}}^2)$, and the computational complexity is reduced to $\mathcal{O}(d_{\text{in}}^3 + d_{\text{out}}^3)$, achieving a compression from product-level complexity to summation-level complexity.

Server-Side Optimization for Global Aggregation

Upon completing the computation of activation covariance matrices $\mathbf{A}_{k,l}^{(e)}$ and gradient covariance matrices $\mathbf{B}_{k,l}^{(e)}$, clients upload the expert layer information to the server for posterior aggregation. For notational clarity, we define $\xi_l^{(e)} = \text{vec}(\mu_l^{(e)})$ as the vectorized global parameter representation of expert e at layer l , and $\zeta_{k,l}^{(e)} = \text{vec}(\mu_{k,l}^{(e)})$ as the vectorized local parameter representation uploaded by client k . Combining Eq. (9) and Eq. (13), the globally optimal parameters are aggregated as:

$$\xi_l^{(e)} = \left(\sum_{k=1}^K \mathbf{B}_{k,l}^{(e)} \otimes \mathbf{A}_{k,l}^{(e)} \right)^{-1} \cdot \sum_{k=1}^K \left[(\mathbf{B}_{k,l}^{(e)} \otimes \mathbf{A}_{k,l}^{(e)}) \zeta_{k,l}^{(e)} \right], \quad (15)$$

To simplify computation, we reshape the local optimal parameter vector $\mu_{k,l}^{(e)}$ back into its corresponding two-dimensional weight matrix form, denoted $\mathbf{W}_{k,l}^{(e)}$. Leveraging the algebraic identity relating the Kronecker product to the vectorization operator, the aggregation can be equivalently expressed as:

$$\text{vec}(\mathbf{W}_l^{(e)}) = \left(\sum_{k=1}^K \mathbf{B}_{k,l}^{(e)} \otimes \mathbf{A}_{k,l}^{(e)} \right)^{-1} \sum_{k=1}^K \text{vec}(\mathbf{B}_{k,l}^{(e)} \mathbf{W}_{k,l}^{(e)} \mathbf{A}_{k,l}^{(e)}). \quad (16)$$

Through this identity transformation, the originally required high-dimensional matrix operations are equivalently reduced to successive multiplications of three low-dimensional matrices, avoiding potential memory overflow during aggregation and compressing computational overhead to standard matrix multiplication levels. Combining with Eq. (8), the global precision matrix is:

$$\mathbf{H}_l^{(e)} = \sum_{k=1}^K \mathbf{B}_{k,l}^{(e)} \otimes \mathbf{A}_{k,l}^{(e)}. \quad (17)$$

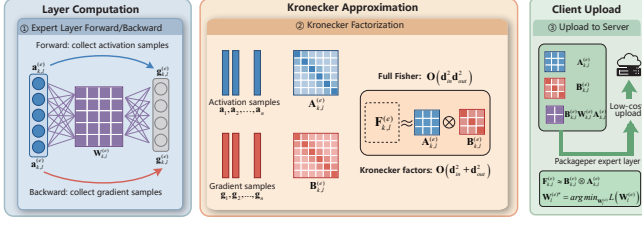


Figure 3: Kronecker approximation of the layer-wise empirical Fisher matrix.

However, Eq. (17) requires inverting the sum of Kronecker products from all clients, which cannot be decomposed into individual factor inverses and must be treated as a full matrix. To avoid this costly matrix inversion, we reformulate global aggregation as a server-side convex optimization problem. Considering that $\mathbf{A}_{k,l}^{(e)}$ and $\mathbf{B}_{k,l}^{(e)}$ are factorized components of the empirical Fisher matrix and are naturally symmetric positive semi-definite, the quadratic objective constructed from them exhibits favorable convexity and optimization stability. Based on this property, we reconstruct the high-dimensional posterior aggregation problem as a server-side convex optimization process, where a quadratic objective is constructed and gradient descent is applied to efficiently search for the globally optimal model parameters. Specifically, let $\mathbf{W}_l^{(e)}$ denote the global weight matrix of layer l in expert e . Instead of directly solving the inverse-form aggregation in Eq. (16), the server optimizes $\mathbf{W}_l^{(e)}$ so that its projected representation matches the sum of client-uploaded local projections, leading to the following quadratic loss:

$$\mathcal{L}(\mathbf{W}_l^{(e)}) = \frac{1}{2} \left\| \sum_{k=1}^K \text{vec} \left(\mathbf{B}_{k,l}^{(e)} \mathbf{W}_l^{(e)} \mathbf{A}_{k,l}^{(e)} \right) - \sum_{k=1}^K \text{vec} \left(\mathbf{B}_{k,l}^{(e)} \mathbf{W}_{k,l}^{(e)} \mathbf{A}_{k,l}^{(e)} \right) \right\|_2^2. \quad (18)$$

The server-side global optimal parameter aggregation is thus transformed into the optimization problem:

$$\mathbf{W}_l^{(e)*} = \arg \min_{\mathbf{W}_l^{(e)}} \mathcal{L}(\mathbf{W}_l^{(e)}). \quad (19)$$

In the practical aggregation phase, the right-hand term $\sum_k \text{vec}(\mathbf{B}_{k,l}^{(e)} \mathbf{W}_{k,l}^{(e)} \mathbf{A}_{k,l}^{(e)})$ in Eq. (18) constitutes a known constant vector assembled entirely from information uploaded by clients. Since both the input covariance matrix $\mathbf{A}_{k,l}^{(e)}$ and the gradient covariance matrix $\mathbf{B}_{k,l}^{(e)}$ are positive semi-definite, $\mathcal{L}(\mathbf{W}_l^{(e)})$ is strictly convex in the parameter space. The server can therefore directly employ standard gradient descent with automatic differentiation to iteratively update $\mathbf{W}_l^{(e)}$, converging to the global optimum at a guaranteed linear rate. This enables efficient layer-wise expert parameter aggregation in FL. The above process is repeated until convergence. For completeness, the detailed pseudo-code of BLA-FedMoE is provided in Algorithm 1.

Algorithm 1: BLA-FedMoE: Layer-Aware Posterior Aggregation for Federated MoE

Require: Number of clients K , number of layers L , number of experts E , server optimization iterations T

- 1: **Initialize:** Global expert weight matrices $\mathbf{W}_l^{(e)}$ for all $l \in \{1, \dots, L\}$, $e \in \{1, \dots, E\}$
 - 2: **Client-side execution** (for each $k \in \{1, \dots, K\}$):
 - 3: Train local MoE model on heterogeneous dataset $\mathcal{D}_k$; obtain expert weights $\mathbf{W}_{k,l}^{(e)}$
 - 4: Perform layer-wise Laplace approximation; compute Kronecker factors $\mathbf{A}_{k,l}^{(e)}$, $\mathbf{B}_{k,l}^{(e)}$
 - 5: Upload $\{\mathbf{A}_{k,l}^{(e)}, \mathbf{B}_{k,l}^{(e)}, \mathbf{B}_{k,l}^{(e)} \mathbf{W}_{k,l}^{(e)} \mathbf{A}_{k,l}^{(e)}\}$ to server
 - 6: **Server-side execution:**
 - 7: **for** $l = 1, \dots, L$ **do**
 - 8: **for** $e = 1, \dots, E$ **do**
 - 9: Construct target vector: $\mathbf{c}_l^{(e)} = \sum_{k=1}^K \text{vec}(\mathbf{B}_{k,l}^{(e)} \mathbf{W}_{k,l}^{(e)} \mathbf{A}_{k,l}^{(e)})$
 - 10: Solve via gradient descent for T iterations according to Eq. (19) to obtain $\mathbf{W}_l^{(e)*}$
 - 11: Assign: $\mathbf{W}_l^{(e)} \leftarrow \mathbf{W}_l^{(e)*}$
 - 12: **end for**
 - 13: **end for**
 - 14: **return** Aggregated global MoE model parameters $\{\mathbf{W}_l^{(e)}\}_{l=1, e=1}^{L, E}$
-

Experiments

In this section, we conduct comprehensive experiments to evaluate the effectiveness of the proposed BLA-FedMoE framework. We first describe the experimental setup, including datasets, baselines, and implementation details. We then present results on test accuracy and ablation studies to validate each component of our method.

Experimental Setup

Datasets and Model Architecture. We evaluate our method on five widely used benchmark datasets under non-IID partitions: CIFAR-10 (Krizhevsky, Hinton et al. 2009; Li et al. 2024), CIFAR-100 (Krizhevsky, Hinton et al. 2009; Lu et al. 2024), Tiny-ImageNet (Le, Yang et al. 2015), CINIC-10 (Darlow et al. 2018), and STL-10 (Coates, Ng, and Lee 2011). CIFAR-10 contains 60,000 32×32 color images from 10 classes, while CIFAR-100 follows the same image format but includes 100 fine-grained classes, making the classification task more challenging. Tiny-ImageNet contains 200 classes with images resized to 64×64 , providing a more complex visual recognition benchmark. CINIC-10 is constructed from CIFAR-10 and downsampled ImageNet images, which increases the data scale and diversity while keeping 10 classes. STL-10 contains higher-resolution natural images from 10 classes and is commonly used to evaluate image classification and representation learning performance. In the main experiments, we evaluate three backbone architectures, including VGG-11, ResNet-18, and ViT-Small, each augmented with a sparse MoE classification head. Specifically, the extracted feature representation is fed into a MoE module consisting of $E = 4$ FFN experts. A top-2 gating mechanism

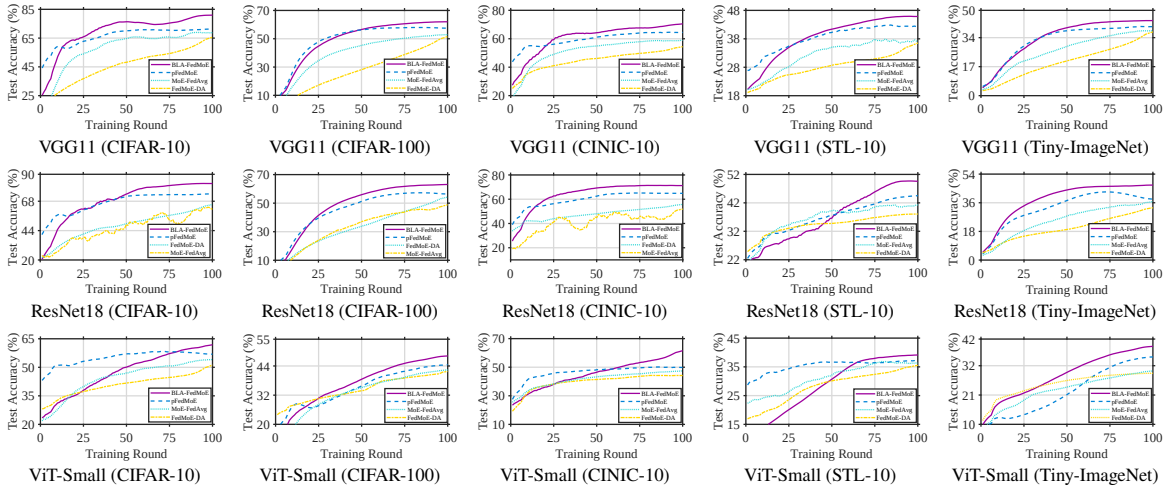


Figure 4: Test accuracy comparison of different backbones across five datasets under $\alpha = 0.1$.

is employed to activate two experts for each input representation.

Non-IID Data Partitioning. To simulate realistic statistical heterogeneity, we employ the Dirichlet distribution-based partitioning strategy (Hsu, Qi, and Brown 2019; Yan et al. 2025). Specifically, for each client k , the proportion of samples from class c is drawn from $\text{Dir}(\alpha)$, where smaller values of α correspond to higher degrees of data skew. We conduct experiments under $\alpha \in \{0.1, 0.3, 0.5\}$ to evaluate performance across varying heterogeneity levels.

Baselines. We compare BLA-FedMoE with three representative baselines. FedAvg (McMahan et al. 2017) is extended to the same MoE architecture to isolate the effect of aggregation. FedMoE-DA (Zhan et al. 2024) selectively aggregates correlated experts with personalized gating networks, while pFedMoE (Yi et al. 2026) combines shared and private feature extractors for personalized federated MoE learning.

Implementation Details. Unless otherwise specified, all experiments are conducted with ResNet-18 as the backbone, where $K = 10$ clients participate in each communication round. Local training runs for 2 epochs per round with a batch size of 64, and the total number of communication rounds is set to 100. For the optimization in BLA-FedMoE, we use the Adam optimizer with a learning rate of 0.001. The damping coefficients are set to $\gamma_A = \gamma_B = 0.01$.

Main Results

Test Accuracy. Figure 4 presents the test accuracy results of all methods with three backbone architectures, including VGG-11 (Simonyan and Zisserman 2015; So et al. 2023), ResNet-18 (He et al. 2016; Li et al. 2020b), and ViT-Small (Touvron et al. 2021), across five datasets: CIFAR-10, CIFAR-100, Tiny-ImageNet, CINIC-10, and STL-10. The following observations can be made: First, BLA-FedMoE consistently achieves the highest test accuracy across different backbones and datasets. Specifically, on VGG-11,

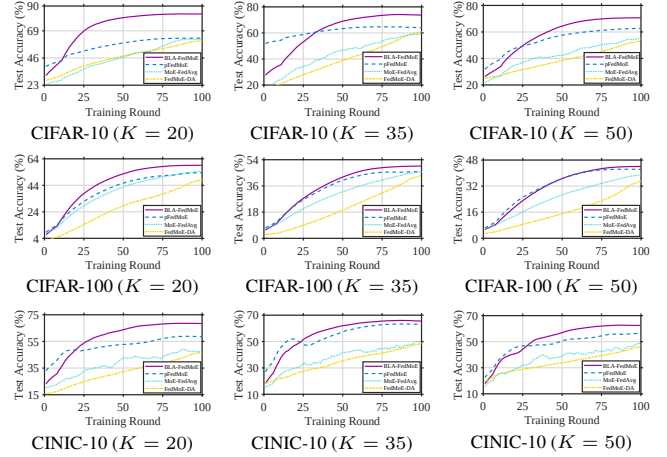


Figure 5: Test accuracy under varying numbers of clients with $\alpha = 0.1$.

ResNet-18, and ViT-Small, BLA-FedMoE outperforms the second-best baseline by 5.6%, 5.7%, and 5.2% percentage points on average, respectively. Second, BLA-FedMoE maintains stable advantages under all three backbone architectures, indicating that the proposed aggregation strategy is not restricted to a specific feature extractor. This suggests that layer-wise posterior-aware aggregation can be effectively integrated with both convolutional and transformer-based architectures. Third, under the fixed highly non-IID setting, BLA-FedMoE remains effective across datasets with different visual complexity. This demonstrates the general applicability of the proposed method to MoE models under heterogeneous FL settings.

Scalability Analysis

Varying Number of Clients. We further analyze the effect of different client scales on model performance and evaluate the scalability of BLA-FedMoE on CIFAR-10, CIFAR-100, and CINIC-10. The non-IID level is fixed at $\alpha = 0.1$, and the

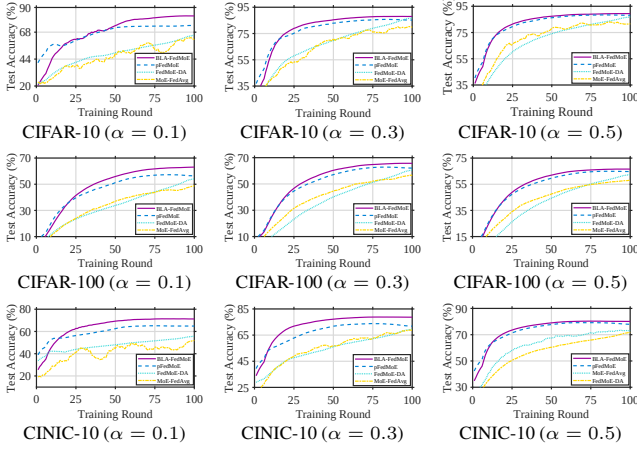


Figure 6: Test accuracy under varying Non-IID levels.

number of clients is set to $K \in \{20, 35, 50\}$. As shown in Figure 5, BLA-FedMoE consistently outperforms baselines across different datasets and client scales. This is because layer-wise posterior confidence guides expert aggregation toward high-certainty parameter directions, thereby alleviating the parameter conflicts caused by more fragmented multi-client data distributions under strong heterogeneity. These results demonstrate the strong scalability and stability of BLA-FedMoE in multi-client federated settings.

Varying Non-IID Levels. We evaluate BLA-FedMoE under different non-IID levels on CIFAR-10, CIFAR-100, and CINIC-10 with $\alpha \in 0.1, 0.3, 0.5$. As shown in Figure 6, BLA-FedMoE achieves performance comparable to existing methods under the mild heterogeneous setting with $\alpha = 0.5$, while its advantage becomes more evident when the data distribution becomes highly heterogeneous, i.e., $\alpha = 0.1$. This trend can be explained by the increased divergence among local expert updates under smaller α , where conventional averaging-based aggregation may mix conflicting parameter directions from different clients and dilute reliable expert knowledge learned from informative local data. In contrast, BLA-FedMoE leverages layer-wise posterior precision to assign larger weights to more trustworthy expert parameters while suppressing uncertain updates induced by biased local distributions. As a result, it provides a more stable and effective aggregation mechanism under severe non-IID data heterogeneity.

Ablation Studies

We evaluate each component of BLA-FedMoE on CIFAR-10 and CIFAR-100 ($\alpha = 0.1, 0.3, 0.5$), with results shown in Figure 7.

Effect of Layer-Wise Posterior Inference. To assess the importance of layer-level granularity, we compare BLA-FedMoE with a variant that uses a single global Fisher matrix averaged across all layers for each expert, denoted as BLA-Global. The ablation study considers three non-IID levels, i.e., $\alpha = 0.1, 0.3, 0.5$. As shown in Figure 7, BLA-Global consistently underperforms BLA-FedMoE, with an

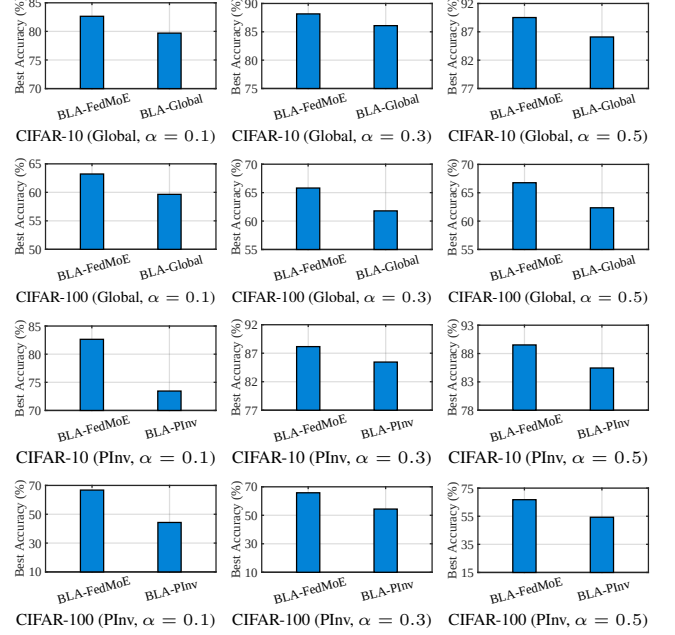


Figure 7: Ablation study: layer-wise posterior and optimization.

average accuracy drop of 2.4% on CIFAR-10 and 3.8% on CIFAR-100. This confirms that different layers exhibit distinct sensitivities to local data and that layer-aware posterior aggregation is essential.

Effect of Optimization. To assess the importance of optimization, we compare BLA-FedMoE with a variant that replaces the original iterative optimization process with a single-step pseudo-inverse approximation, denoted as BLA-PInv. The ablation study considers three non-IID levels, i.e., $\alpha = 0.1, 0.3, 0.5$. As shown in Figure 7, BLA-PInv consistently underperforms BLA-FedMoE, with an average accuracy drop of 4.1% on CIFAR-10 and 15.4% on CIFAR-100. This is mainly because the pseudo-inverse solution is less stable, and such a single-step approximation may amplify noisy or low-confidence directions, leading to inaccurate global expert parameters. In contrast, the proposed iterative optimization method progressively minimizes the aggregation objective and provides a more stable solution, demonstrating its necessity for reliable expert aggregation.

Conclusion

In this paper, we presented BLA-FedMoE, a layer-aware posterior aggregation framework for MoE models under non-IID data distributions in FL. By modeling expert aggregation from a Bayesian posterior perspective, BLA-FedMoE estimates layer-wise parameter certainty with block-diagonal empirical Fisher information matrices and solves global expert parameters through server-side convex optimization. Extensive experiments show that BLA-FedMoE consistently outperforms existing federated MoE baselines, demonstrating its effectiveness in preserving expert specificity and improving generalization under heterogeneous data.

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